

Pranav Guhathakurta

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SUMMARY

Software engineer with hands-on experience building cloud-native compliance and fintech systems. Delivered production-grade microservices and dashboards used by sovereign clients and SME users. Skilled in backend development (Node.js, AWS, GCP), ML prototyping, and full-stack deployment. Actively seeking new grad SWE roles in infrastructure, data, or backend-focused teams.

EDUCATION

University of Pennsylvania, The Wharton School - Philadelphia, PA

August 2022 - May 2026

Bachelor of Science (B.S.) in Economics

Minor: Computer Science | GPA: 3.55/4.00

Honors: Joseph Wharton Scholar, Benjamin Franklin Scholar

Relevant Coursework: Data Structures & Algorithms, Programming, Multivariable Calculus, Quantitative Modeling, Business Ethics

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, HTML/CSS

Frameworks & Libraries: Node.js, React, Express, Micronaut, MongoDB, DynamoDB

Cloud & DevOps: AWS (Lambda, S3, API Gateway, Cognito), Firebase, Docker, Git, GitHub Actions, Linux CLI, GCP (Cloud Functions, Pub/Sub)

Databases: PostgreSQL, SQL Server, MongoDB, DynamoDB

Testing & CI: Jest, JUnit, Pytest, GitHub Actions

Concepts: REST APIs, JSON, HTTP, CI/CD, Microservices, Event-Driven Architecture

CERTIFICATIONS

AWS Certified Cloud Practitioner - Amazon Web Services (September 2023 - September 2026)

Tools for Data Science - IBM via Coursera (June 2021)

Fundamentals of Quantitative Modeling - Wharton Online via Coursera (June 2022)

EXPERIENCE

Flagright (Y Combinator S22) - San Francisco, CA

May 2024 - August 2024

Software Engineer, Internship

AI-native AML compliance startup that provides fintechs and banks with real-time transaction monitoring, risk scoring, and case management via API-first infrastructure.

- Led deployment of a Node.js microservice for real-time sanctions screening; supported 25K+ transactions/day via AWS API Gateway with <800 ms response latency.
- Designed SOC-2 audit-readiness framework and authored a Business Continuity Plan across 5+ production services; aligned compliance architecture with fintech standards.
- Evaluated LLM-powered vendors for GenAI document analysis; led benchmarking and co-authored a whitepaper on AML/KYC enhancements.
- Participated onsite at Kuala Lumpur for on-prem deployment for a Malaysian sovereign fund, tailoring compliance workflows to integrate with their infrastructure.

Iridium Business Solutions - Cape Town, South Africa
Software Engineer, Internship

May 2023 - July 2023

Cloud-based accounting, advisory, and automation firm helping SMEs streamline their finance operations via tech-driven automation tools.

- Developed a financial analytics dashboard using React + AWS (CloudFront, S3, DynamoDB, Lambda, Cognito, API Gateway) for 50+ SME clients.
- Built 10+ REST endpoints and secure UI components managing 1K+ daily requests; implemented Cognito-based role authentication.
- Streamlined reporting workflows in accounting/payroll SaaS integration, improving client throughput by 30%.

FinTech Global - New York, NY
Event & Product Operations Coordinator (Contract)

September 2023, September 2024

RegTech/InsureTech event production and networking support in fintech industry forums.

- Coordinated logistics and stage operations at industry events, introduced keynote speakers, and organized technical networking sessions for participants.

PROJECTS & RESEARCH

Sanctions Filtering with AI - Joseph Wharton Scholars Research

Node.js, Express, MongoDB, Python, NLP

Designed and prototyped a system for identifying transactions involving politically exposed persons and OFAC-sanctioned entities.

- Built ingestion pipeline for structured/unstructured data; integrated graph traversal logic for link-based risk detection.
- Delivered real-time alerting with optimized response time (P95 latency target under 1s); evaluated early-stage model precision.

HybridGraphDB - Personal Project

Python, Faiss, FastAPI, Docker, MongoDB

Created a hybrid graph-document database capable of vector similarity routing.

- Used HNSW index and node embeddings to optimize graph traversal queries across financial datasets.
- Achieved 92% top-K recall and reduced end-to-end query time by 38% in benchmark tests.

Twitter Stock Price Predictor - Personal Project

Python, Tweepy, Tiingo API, scikit-learn, textBlob

Scraped and analyzed tweet sentiment data to predict short-term stock price movements.

- Built SVM model trained on tweet clusters; achieved 80% directional accuracy on test tickers.

Othello AI - Columbia SHAPE Program

Python, Heuristic Search, Object-Oriented Programming

Built a competitive Othello/Reversi AI agent using Minimax and alpha-beta pruning.

- Won head-to-head challenge at Columbia SHAPE; optimized heuristic weights through tournament feedback loop.